

Window Function

Lab 1: Film Ranking

```
WITH RankedFilms AS (  
  SELECT  
    c.name AS category_name,  
    f.title AS film_title,  
    f.length,  
    DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC, f.title) AS  
length_rank  
  FROM film f  
  JOIN film_category fc ON f.film_id = fc.film_id  
  JOIN category c ON fc.category_id = c.category_id  
)  
SELECT category_name, film_title, length, length_rank  
FROM RankedFilms  
WHERE length_rank <= 3  
ORDER BY  
  category_name,  
  length_rank,  
  film_title;
```

category_name character varying (25) 🔒	film_title character varying (255) 🔒	length smallint 🔒	length_rank bigint 🔒
Action	Darn Forrester	185	1
Action	Worst Banger	185	2
Action	Casualties Encino	179	3
Animation	Gangs Pride	185	1
Animation	Pond Seattle	185	2
Animation	Sons Interview	184	3
Classics	Conspiracy Spirit	184	1
Classics	Dracula Crystal	176	2
Classics	Lights Deer	174	3
Comedy	Control Anthem	185	1
Comedy	Saturn Name	182	2
Comedy	Searchers Wait	182	3
Children	Fury Murder	178	1
Children	Wrong Behavior	178	2
Children	Empire Malkovich	177	3
Documentary	Wife Turn	183	1
Documentary	Young Language	183	2
Documentary	Cause Date	179	3

Lab 2: Growth Analysis

```
WITH MonthlyRevenue AS (  
    SELECT DATE_TRUNC('month', payment_date) AS revenue_month, SUM(amount) AS  
    current_revenue  
    FROM payment  
    GROUP BY DATE_TRUNC('month', payment_date)  
),  
RevenueWithLag AS (  
    SELECT revenue_month, current_revenue, LAG(current_revenue) OVER (ORDER BY  
    revenue_month) AS prev_revenue  
    FROM MonthlyRevenue  
)  
SELECT revenue_month,  
    current_revenue,  
    prev_revenue,  
    ROUND(((current_revenue - prev_revenue) / NULLIF(prev_revenue, 0)) * 100, 2) AS  
mom_growth_percentage  
FROM RevenueWithLag  
ORDER BY revenue_month;
```

revenue_month timestamp without time zone 🔒	current_revenue numeric 🔒	prev_revenue numeric 🔒	mom_growth_percentage numeric 🔒
2007-02-01 00:00:00	8351.84	[null]	[null]
2007-03-01 00:00:00	23886.56	8351.84	186.00
2007-04-01 00:00:00	28559.46	23886.56	19.56
2007-05-01 00:00:00	514.18	28559.46	-98.20

Lab 3: Customer Lifetime Value

```
SELECT customer_id, payment_id, payment_date, amount, SUM(amount) OVER  
(PARTITION BY customer_id ORDER BY payment_date, payment_id) AS cumulative_clv  
FROM payment  
ORDER BY customer_id, payment_date;
```

customer_id smallint	payment_id [PK] integer	payment_date timestamp without time zone	amount numeric (5,2)	cumulative_clv numeric
1	18495	2007-02-14 23:22:38.996577	5.99	5.99
1	18496	2007-02-15 16:31:19.996577	0.99	6.98
1	18497	2007-02-15 19:37:12.996577	9.99	16.97
1	18498	2007-02-16 13:47:23.996577	4.99	21.96
1	18499	2007-02-18 07:10:14.996577	4.99	26.95
1	18500	2007-02-18 12:02:25.996577	0.99	27.94
1	18501	2007-02-21 04:53:11.996577	3.99	31.93
1	22680	2007-03-01 07:19:30.996577	4.99	36.92
1	22681	2007-03-02 14:05:18.996577	3.99	40.91
1	22682	2007-03-02 16:30:04.996577	0.99	41.90
1	22683	2007-03-17 11:06:20.996577	4.99	46.89
1	22684	2007-03-18 02:25:55.996577	0.99	47.88
1	22685	2007-03-19 08:23:42.996577	0.99	48.87
1	22686	2007-03-19 12:25:20.996577	2.99	51.86
1	22687	2007-03-21 22:02:23.996577	0.99	52.85
1	22688	2007-03-21 23:56:23.996577	1.99	54.84
1	22689	2007-03-22 18:10:03.996577	2.99	57.83
1	22690	2007-03-22 18:32:12.996577	5.99	63.82

Lab 4: The “Top-N” Filter Challenge

```
WITH rev_rank AS(
    SELECT c.customer_id, c.first_name, c.last_name, co.country, SUM(p.amount) AS
total_spend, RANK() OVER (PARTITION BY co.country ORDER BY SUM(p.amount) DESC)
AS spending_rank
    FROM customer c
    JOIN address a ON c.address_id = a.address_id
    JOIN city ci ON a.city_id = ci.city_id
    JOIN country co ON co.country_id = ci.country_id
    JOIN payment p ON p.customer_id = c.customer_id
    GROUP BY c.customer_id, co.country
)
```

```
SELECT country,customer_id,first_name,last_name,total_spend
FROM rev_rank
WHERE spending_rank = 1
ORDER BY country;
```

country character varying (50)	customer_id integer	first_name character varying (45)	last_name character varying (45)	total_spend numeric
Afghanistan	218	Vera	Mccoy	67.82
Algeria	176	June	Carroll	151.68
American Samoa	320	Anthony	Schwab	47.85
Angola	528	Claude	Herzog	93.80
Anguilla	381	Bobby	Boudreau	99.68
Argentina	560	Jordan	Archuleta	129.71
Armenia	41	Stephanie	Mitchell	118.75
Austria	266	Nora	Herrera	113.74
Azerbaijan	334	Raymond	Mcwhorter	108.75
Bahrain	590	Seth	Hannon	108.76
Bangladesh	21	Michelle	Clark	146.68
Belarus	144	Clara	Shaw	189.60
Bolivia	431	Joel	Francisco	92.78
Brazil	178	Marion	Snyder	194.61
Brunei	91	Lois	Butler	107.66
Bulgaria	540	Tyrone	Asher	112.76
Cambodia	516	Elmer	Noe	92.76
Cameroon	352	Albert	Crouse	96.78